ModelIt!

Student Activity Pack

Why Do Some Things Dissolve and Others Don't?

How Water Temperature and Motion Affect Dissolving Speed

Grade 05L2 | NGSS 5-PS1-1, 5-PS1-4 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What does it mean when something 'dissolves'? Where does the solid go?

2 If you stir sugar into water, is the sugar still there? How do you know?

3 Draw what you think is happening to the sugar particles when sugar dissolves in water.

4 What could you do to make something dissolve faster?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Water Temperature | Stirring Speed | Particle Contact Rate | Dissolving Speed | Solution Concentration

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Hot Water, Fast Stirring

High temperature, fast stirring

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Cold Water, No Stirring

Low temperature, no stirring

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Hot Water, No Stirring

High temperature, no stirring

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Particle Size — Smaller pieces have more surface area exposed to water, dissolving faster

Solvent Amount — More water can dissolve more substance before reaching saturation

Substance Type — Different substances have different solubilities — salt dissolves but sand does not

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do some things dissolve in water quickly while others take forever or don't dissolve at all?
Use evidence from your simulation to support your answers.

1 When sugar dissolves, does it disappear? How do you know it is still there?

2 Why does stirring help even if the water temperature stays the same?

3 What surprised you most about how temperature affected dissolving speed?

4 If you could add one more component to this model, what would it be and why?

5 How does understanding dissolving help in cooking, medicine, or manufacturing?

8 STEM Challenge: Design the Fastest Dissolving Method

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YOUR ENGINEERING MISSION

Design and test the fastest method to completely dissolve a sugar cube in water.

THE SCENARIO

A drink company needs to dissolve flavor packets as fast as possible in their factory. They can adjust water temperature, stirring method, and other factors. Your team must find the fastest dissolving method and explain the science behind it.

Engineering Design Thinking

1. WHAT WATER TEMPERATURE WILL YOU USE AND WHY?
2. WHAT STIRRING METHOD WILL YOU USE (SPOON, SHAKING, BLENDING)?
3. WILL YOU CHANGE THE SHAPE OR SIZE OF THE SUGAR CUBE? WHY?
4. HOW WILL YOU MEASURE THE EXACT TIME IT TAKES TO FULLY DISSOLVE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Food Scientists and Chemical Engineers design processes for mixing, dissolving, and creating solutions at industrial scale. They use the same principles you modeled today to make medicines, beverages, and materials. They earn $70,000-$125,000/year.